The pair singular series of a polynomial II: zeros of Dedekind zeta functions in the second moment of prime values

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Abstract

How much does the number of primes among the values $f(1), f(2), \ldots$ of a polynomial fluctuate? Part I of this work showed that the answer is encoded in an explicit arithmetic function, the pair singular series $S_f(h)$, and that the diagonal part of its second moment has the Dedekind zeta function ζ_K of $K = \mathbb{Q}[t]/(f)$ as generating function. Here we determine that generating function completely. It is an infinite product of Artin L-functions of the Galois group of f, evaluated at $N(s+1)$, $N = 1, 2, 3, \ldots$, whose exponents are virtual characters given by a plethystic logarithm; for quadratics the exponents are necklace numbers. The first factor beyond ζ_K is $1/L(2s+2, \mathrm{Sym}^2 V)$, and $L(s, \mathrm{Sym}^2 V)$ is the product of the Dedekind zeta functions of the field of a root and of the field of a pair of roots. The line $\Re s = -1$ is a natural boundary, for every f including the twin-prime case. Consequently the Riesz means of the diagonal satisfy an explicit formula whose oscillating part is indexed by the zeros of these Dedekind zeta functions, with terms $x^{m-1+\rho/2}$, and whose smooth part contains an unexpected term $x^{m-2/3}$; the formula is proved for quadratics under the Riemann hypothesis for ζ and $L(s, \chi_D)$, and for general f under the corresponding hypotheses; unconditionally the error term is $\Omega_\pm(x^{m-3/4})$. For $f = t^2 + 1$ every coefficient is computed and the formula is verified against the exact Riesz means to $x = 4 \cdot 10^7$: the data reproduce the predicted sum over the zeros of ζ and $L(s, \chi_{-4})$ with regression coefficient 1.008. The remaining conjecture of part I (the off-diagonal) is reformulated as the analytic continuation of a Dirichlet series of Weyl sums of the roots of f, for quadratics a series of Salié-type sums, and the two estimates its proof requires are stated. Assuming the Hardy–Littlewood pair conjecture these results describe the second-order term of the variance of the number of prime values of f; the zero-induced fluctuations lie below the scale at which that transfer is valid, and no statement about prime values themselves and the zeros of ζ_K is claimed.

1 Introduction

The question. Let $f \in \mathbb{Z}[t]$ be irreducible, without fixed prime divisor, and let $C(f)$ be its Bateman–Horn constant, the conjectural density factor for primes among $f(1), f(2), f(3), \ldots$. Part I of this work [35] studied not the density but the *fluctuation* of the number of primes along the sequence, through the pair singular series

$$S_f(h) = \prod_p \frac{1 - \omega_{f,2}(p,h)/p}{(1 - 1/p)^2}, \qquad \omega_{f,2}(p,h) = 2\omega_f(p) - \nu_f(p,h),$$

where $\omega_f(p)$ is the number of roots of f modulo p and $\nu_f(p,h)$ the number of roots s with $s + h$ again a root. Under the Hardy–Littlewood conjecture $S_f(h)/(\log f(t) \log f(t+h))$ is the probability that $f(t)$ and $f(t+h)$ are both prime, and the quantity that measures how far the primes along f are from independent trials is

$$\Sigma_f(H) = \sum_{h \le H} \big(S_f(h) - C(f)^2\big). \tag{1}$$

Part I proved an exact identity,

$$\Sigma_f(H) = -C^2 \sum_{d\geq 2} a_f(d)\Big\{\frac{H}{d}\Big\} + C^2\,\mathrm{Off}_f(H), \qquad a_f(d) = W_f(d)\,\omega_f(d), \tag{2}$$

with explicit multiplicative weights $W_f(d)$ on squarefree d (notation below), where the first sum, the *diagonal*, comes from pairs of equal roots of f modulo d and Off_f, the *off-diagonal*, from pairs of distinct roots; it showed that the diagonal has Dirichlet series

$$D_f(s) = \sum_d a_f(d)d^{-s} = \zeta_K(s+1)E_f(s), \qquad K = \mathbb{Q}[t]/(f), \qquad \operatorname*{Res}_{s=0} D_f = \frac{1}{C(f)};$$

and it conjectured (Conjecture 1 there)

$$\Sigma_f(H) = -\tfrac{1}{2}\,C(f)\log H + A_f + o(1), \tag{3}$$

a theorem for linear f (Friedlander–Goldston, Vaughan, Montgomery–Soundararajan) and equivalent, in Cesàro form, to the statement that the off-diagonal contributes no logarithm ("Hypothesis (E)"). This paper determines E_f and draws the consequences; the off-diagonal is reformulated in Section 5 and left for part III.

In plain terms. The primes along f are not independent trials: divisibility of $f(t)$ by a small prime p is periodic in t, so the small primes act deterministically and only the large ones are random. $\Sigma_f(H)$ is the resulting total correlation up to distance H, and its main term $-\frac{1}{2}C(f)\log H$ is the polynomial version of the fact that primes in short intervals are more regular than random. This paper is about the next layer, the oscillations of Σ_f around its main terms. For the twin-prime problem Goldston and Suriajaya showed that these oscillations are a sum of waves, one for each zero of the Riemann zeta function. For a polynomial the same happens with the zeros of the Dedekind zeta functions of the number fields cut out by f; there is one more smooth term, $x^{m-2/3}$, that the linear case does not have; the generating function has a natural boundary at $\Re s = -1$, which changes what can be proved; and all of this is visible in exact computations for t^2+1. Nothing here is a statement about primes themselves: that needs the Hardy–Littlewood pair conjecture, and the conjecture (3) remains a conjecture, now reduced to a precise analytic statement.

The linear case as template. For $f(t) = t$, $S_f = \mathfrak{S}$ is the twin-prime singular series, and Goldston and Suriajaya [17] proved that the Riesz means $\sum_{k\leq x}(x-k)^m\mathfrak{S}(k)$, $m \geq 2$, equal $x^{m+1}/(m+1) - \frac{1}{2}x^m(\log x - H_m + \gamma + \log 2\pi) + x^{m-1}\sum_{|\gamma|\leq U} a(\rho)x^{\rho/2} + O(x^{m-1+\varepsilon})$, the zeros of ζ entering through the factor $1/\zeta(2s+2)$ of the generating series, with error $\Omega_\pm(x^{m-3/4})$ (see also [16, 39, 14]). Our generating series is $\zeta_K(s+1)E_f(s)$; the question is what E_f is.

Results. Section 2: E_f is an infinite product of Artin L-functions $L(N(s+1),\Psi_N)$, $N \geq 2$, with virtual characters Ψ_N given by an explicit formula (Theorem 4); $\Psi_2 = -\mathrm{Sym}^2 V$ and $L(s,\mathrm{Sym}^2 V)$ is a product of Dedekind zeta functions of "pair fields" (Lemma 1); for quadratics the exponents are necklace numbers (Proposition 6); $\Re s = -1$ is a natural boundary (Theorem 7). Section 3: the explicit formula for the Riesz means of the diagonal, for quadratics under the Riemann hypothesis for ζ and $L(s,\chi_D)$ with a complete proof (Theorem 10) and for general f under the analogous hypotheses (Theorem 14), and its comparison with exact data for t^2+1. Section 4: unconditionally the error term is $\Omega_\pm(x^{m-3/4})$ (Theorem 15). Section 5: Hypothesis (E) as the analytic continuation of a Dirichlet series of Weyl sums of roots, and the two estimates that would prove it for quadratics. Section 6: what carries over to $\mathbb{F}_q[u]$. Section 7: what all of this means for prime values under the Hardy–Littlewood conjecture. Section 8: what should be done next. Appendix A specifies the computations.

What is conditional on what. As in part I: (A) Bateman–Horn for non-linear f, open, used nowhere; (B) the Hardy–Littlewood pair conjecture for f, used only in Section 7; (C) Conjecture (3), equivalently Hypothesis (E), a statement about the explicit function S_f, reformulated in Section 5, not proved here; and now (D) the Riemann hypothesis for ζ, $L(s, \chi_D)$ and, for general f, the Artin L-functions of its Galois group, needed for the explicit formulas of Section 3 because the natural boundary prevents the contour from leaving the critical strip. Everything in Section 2 and Theorem 15 is unconditional. The zero-induced fluctuations have relative size $T^{-1/4}$, below the error tolerated by the transfer (B): unconditionally, the low-lying zeros of ζ_K govern the second-order fluctuations of the Riesz means of S_f; under GRH the variance of primes is related to the pair correlation of the zeros of ζ_K by the Goldston–Montgomery theorem [15] in the Selberg-class form of Bui, Keating and Smith [7]; we claim no unconditional link between prime values of f and zeros of ζ_K.

Relation to the literature. The only prior appearance of $\zeta_K(s+1)$ in a singular-series average that we know of is Kuperberg, Rodgers and Roditty-Gershon [27], for prime *elements* of a number field; explicit formulas over the zeros of several L-functions occur in the Goldbach setting [4, 18]; variances governed by higher-degree L-functions over $\mathbb{F}_q[t]$ are in [19]. Natural boundaries of Euler products go back to Estermann [13] and Dahlquist [9], with the multivariable case in [3] and the "ghost" formalism of du Sautoy and Grunewald [10]; the factorisation of Euler products with Frobenian coefficients through L-functions of virtual characters, and the criterion for meromorphy in terms of the representation ring (continuation to the whole plane if and only if the local polynomial is cyclotomic in $R(G)$, otherwise a natural boundary), are due to Kurokawa [28] and Moroz [32]; both, together with the "log factorisation" that produces the virtual characters, are set out in Alberts's survey [2], whose class of Euler products is $\prod_p Q_p(p^{-s})$ with $p \mapsto Q_p$ Frobenian. Our local factor F_p is not of that form: it depends on p^{-1} as well as on p^{-s} and Frob_p, and the two variables do not separate. Relative to that theory, what is new here is this two-variable situation, the explicit characters Ψ_N for the specific series D_f, the location of the boundary, and its consequences for the explicit formula; a general natural-boundary criterion for products $\prod_p W(\mathrm{Frob}_p; p^{-1}, p^{-s})$ presumably follows from Kurokawa–Moroz by the same accumulation argument, but we state only what we prove; we found no previous identification of $\mathrm{Sym}^2 V$ or of a natural boundary in a singular-series average, including Goldston–Suriajaya's series for $f(t) = t$, whose factor $G(s)$ is stated there as continuing to $\Re s > -1$ with no remark on the boundary.

Reader's guide. The theorems are in Sections 2, 3 and 4; each is followed by a short paragraph "in words". Section 5 is the foundation of part III. Appendix A lists every computation with inputs, outputs and cost.

Notation. $f \in \mathbb{Z}[t]$ irreducible of degree n without fixed prime divisor; $G = \mathrm{Gal}(f)$; $V = V_f$ its permutation representation on the roots, $\chi_V(g)$ the number of roots fixed by g; $\omega_f(p) = \chi_V(\mathrm{Frob}_p)$ for unramified p; $K = \mathbb{Q}[t]/(f)$; $w = s+1$; $u = 1/p$, $v = p^{-w}$; $P_p = 1 - \omega_f(p)^2/(p - \omega_f(p))^2$, $b(p) = (p - \omega_f(p))^{-2}$, $W_f(d) = d\, b(d) \prod_{p \nmid d} P_p$ for squarefree d; $a_f = W_f \omega_f$; $A_f = a_f * 1$; $D_f(s) = \sum_d a_f(d) d^{-s}$; $R_m(x) = \sum_{n \le x} (x-n)^m A_f(n)$; $\rho = \beta + i\gamma$ a zero of the L-function in question; for a virtual character Ψ, $L(s, \Psi)$ is its Artin L-function.

2 The analytic structure of D_f

Throughout, $f \in \mathbb{Z}[t]$ is irreducible of degree n, without fixed prime divisor, $G = \mathrm{Gal}(f)$, $V = V_f$ the permutation representation of G on the roots, $K = \mathbb{Q}[t]/(f)$, and for a prime p unramified in the splitting field, $g_p = \mathrm{Frob}_p$ (a conjugacy class) with $\omega_f(p) = \chi_V(g_p)$ fixed points. We write

$w = s+1$, $u = u_p = p^{-1}$, $v = v_p = p^{-w}$. By part I, Theorem 3, $D_f(s) = \prod_p F_p(s)$ with

$$F_p(s) = P_p + p\, b(p)\, \omega_f(p)\, p^{-s} = \frac{1 - 2\omega_f(p)u + \omega_f(p)v}{(1-\omega_f(p)u)^2} \qquad (\omega_f(p) < p), \tag{4}$$

as one checks by writing $P_p = 1 - \omega^2u^2/(1-\omega u)^2$ and $p\, b(p)\omega p^{-s} = \omega v/(1-\omega u)^2$. The point of this section is that $F_p(s) \to 1 + \omega_f(p)v$ as $u \to 0$ with v fixed, and that $1+\omega_f(p)v = 1+\chi_V(g_p)v$ is *not* a finite product of Euler factors of Artin L-functions unless $\chi_V \le 1$, i.e. unless f is linear.

2.1 The pair fields

Lemma 1 (pair fields)**.** $\mathrm{Sym}^2 V \cong V \oplus V^{(2)}$*, where* $V^{(2)}$ *is the permutation representation of* G *on unordered pairs of distinct roots. Consequently*

$$L(s, \mathrm{Sym}^2 V) = \zeta_K(s) \prod_O \zeta_{K_O}(s),$$

the product over the G*-orbits* O *on unordered pairs,* K_O *being the fixed field of the stabiliser of a pair in* O*. If* G *is* 2*-transitive there is one orbit, and* $K^{(2)} = \mathbb{Q}(\alpha+\beta, \alpha\beta)$ *has degree* $n(n-1)/2$*.*

Proof. $\mathrm{Sym}^2 V$ has the basis $e_\alpha e_\beta$ ($\alpha \le \beta$ in any fixed order of the roots), which G permutes; the vectors e_α^2 span a copy of V and the vectors $e_\alpha e_\beta$, $\alpha \ne \beta$, span $V^{(2)}$. A permutation representation $\mathrm{Ind}_H^G \mathbf{1}$ has Artin L-function ζ_{L^H}, L the splitting field; the stabiliser of the pair $\{\alpha, \beta\}$ fixes $\alpha+\beta$ and $\alpha\beta$, and $\mathbb{Q}(\alpha+\beta, \alpha\beta)$ has the right degree when G is 2-transitive. □

For quadratics, $\mathrm{Sym}^2 V = \mathbf{1} \oplus \mathbf{1} \oplus \chi_D$ and $L(s, \mathrm{Sym}^2 V) = \zeta(s)^2 L(s, \chi_D)$; for an S_3-cubic $K^{(2)} \cong K$ and $L(s, \mathrm{Sym}^2 V) = \zeta_K(s)^2$; for an S_4- or A_4-quartic $K^{(2)}$ is the resolvent sextic field.

In words. The symmetric square of the permutation representation, identified in part I as the source of the zeros, is the permutation action on *pairs* of roots. Every zero that appears below is therefore a zero of the Dedekind zeta function of an explicit number field: the field of one root, or the field of a pair of roots.

2.2 The plethystic exponents

Definition 2. *For* $N \ge 1$ *and* $g \in G$ *put*

$$\Psi_N(g) = \frac{1}{N} \sum_{j \mid N} \mu(j)\, (-1)^{N/j+1}\, \chi_V(g^j)^{N/j}.$$

Lemma 3. *(i) Each* Ψ_N *is a virtual character of* G*, and* $\Psi_1 = \chi_V$*. (ii) For every* $g \in G$*, as formal power series in* v*,*

$$1 + \chi_V(g)\, v = \prod_{N \ge 1} \exp\Bigl(\sum_{j\ge1} \frac{\Psi_N(g^j)}{j} v^{Nj}\Bigr) = \prod_{N\ge1} \det\bigl(1 - g\, v^N \mid \Psi_N\bigr)^{-1},$$

where for a virtual character $\Psi = A - B$ *we write* $\det(1-gX \mid \Psi)^{-1} = \det(1-gX \mid A)^{-1} \det(1-gX \mid B)$*. (iii)* $|\Psi_N(g)| \le 2n^N/N$*.*

Proof. (ii) Taking logarithms, the claim is $\sum_{M\ge1} c_M(g)v^M = \sum_{N,j} \Psi_N(g^j)v^{Nj}/j$ with $c_M(g) = (-1)^{M+1}\chi_V(g)^M/M$, i.e. $c_M(g) = \sum_{j\mid M} \Psi_{M/j}(g^j)/j$ for all M and g. Insert the definition: $\sum_{j\mid M} \frac{1}{j} \cdot \frac{j}{M} \sum_{i\mid M/j} \mu(i)(-1)^{M/(ij)+1}\chi_V(g^{ij})^{M/(ij)} = \frac{1}{M}\sum_{k\mid M}(-1)^{M/k+1}\chi_V(g^k)^{M/k} \sum_{i\mid k}\mu(i) = c_M(g)$, since only $k = 1$ survives. (i) Write $R(g,v) = 1+\chi_V(g)v$; its coefficients lie in the representation ring $R(G)$. Suppose $\Psi_1, \ldots, \Psi_{K-1}$ are virtual characters. The series $R(g,v)\prod_{N<K}\det(1 - gv^N \mid \Psi_N)$ has coefficients in $R(G)$ (products of the series $\det(1-gX \mid A)^{\pm1}$, whose coefficients are $\pm$ the characters of exterior or symmetric powers of A), and by (ii) it equals $1 + \Psi_K(g)v^K + O(v^{K+1})$; so $\Psi_K \in R(G)$. The case $N = 1$ is the definition. (iii) $|\chi_V(g^j)| \le n$, and $\sum_{j\mid N} n^{N/j} \le n^N + Nn^{N/2} \le 2n^N$ for $n \ge 2$; for $n = 1$ the claim is trivial. □

For f linear, $\chi_V = 1$ and $1+v = (1-v^2)/(1-v)$: $\Psi_1 = 1$, $\Psi_2 = -1$, $\Psi_N = 0$ for $N \geq 3$, which is the factor $\zeta(s+1)/\zeta(2s+2)$ of the twin-prime case [17]. For $n \geq 2$ infinitely many Ψ_N are non-zero, because $1 + \chi_V(g)v$ with $\chi_V(g) \geq 2$ is not a finite product of cyclotomic polynomials in v.

In words. Let $p \to \infty$ in the local factor of D_f keeping $v = p^{-(s+1)}$ fixed: what remains is $1 + \omega v$, ω the number of roots of f modulo p. For $\omega = 1$ this is a ratio of two cyclotomic polynomials, which is why one factor $1/\zeta(2s+2)$ suffices in the twin-prime problem; for $\omega \geq 2$ no finite product of cyclotomic polynomials equals $1 + \omega v$, and infinitely many L-functions are needed, one for each Ψ_N. Lemma 3 says the Ψ_N are (virtual) characters, so these are Artin L-functions.

2.3 The structure theorem

For a virtual character Ψ of G let $L(s, \Psi)$ be its Artin L-function (a quotient of two Artin L-functions of genuine representations, hence meromorphic in $\mathbb{C}$), with local factor $L_p(s, \Psi) = \det(1 - g_p p^{-s} \mid \Psi)^{-1}$ at unramified p; and let S be the finite set of primes ramified in the splitting field or dividing the leading coefficient of f.

Theorem 4 (structure of D_f)**.** *For every $K \geq 1$,*

$$D_f(s) = \prod_{N=1}^{K} L\big(N(s+1), \Psi_N\big) \cdot M_{f,K}(s), \qquad M_{f,K}(s) = \prod_p F_p(s) \prod_{N=1}^{K} L_p\big(N(s+1), \Psi_N\big)^{-1}, \tag{5}$$

where the Euler product $M_{f,K}$ converges absolutely and locally uniformly in $\Re s > -1 + 1/(K+1)$, so that $M_{f,K}$ is holomorphic there. In particular $L(s+1, \Psi_1) = \zeta_K(s+1)$, so that $E_f = \prod_{N=2}^{K} L(Nw, \Psi_N)\, M_{f,K}$; D_f is meromorphic in $\Re s > -1$; and in $\Re s > -1 + 1/(K+1)$ its poles are among the poles of $\prod_{N \leq K} L(Nw, \Psi_N)$, while its zeros that are not zeros of that product are exactly the zeros of the local factors F_p, $p \notin S$, together with those of the finitely many local factors at $p \in S$.

Proof. Fix K and a compact set in $\Re w > 1/(K+1)$, on which $\Re w \geq \eta > 1/(K+1)$. For $p \notin S$ the local factor of $M_{f,K}$ is, by Lemma 3(ii),

$$M_p(s) = \frac{F_p(s)}{1 + \omega v} \cdot \frac{1 + \omega v}{\prod_{N \leq K} \det(1 - g_p v^N \mid \Psi_N)^{-1}} = \frac{1 - 2\omega u/(1 + \omega v)}{(1 - \omega u)^2} \cdot \exp\Big(- \sum_{M > K} d_M(g_p) v^M\Big),$$

with $\omega = \omega_f(p)$ and $d_M(g) = \sum_{N \mid M,\ N > K} \Psi_N(g^{M/N}) N/M$, so $|d_M| \leq 2n^M \tau(M)$ by Lemma 3(iii). For p so large that $n|v| \leq np^{-\eta} \leq \frac{1}{2}$ and $\omega u \leq \frac{1}{4}$ we get $\big|\sum_{M>K} d_M v^M\big| \leq \sum_{M>K} 2\tau(M)(n|v|)^M \ll_K (n|v|)^{K+1} \ll p^{-(K+1)\eta}$, and $\frac{1-2\omega u/(1+\omega v)}{(1-\omega u)^2} = 1 + \frac{2\omega^2 uv}{1+\omega v} + O(u^2) = 1 + O(p^{-1-\eta}) + O(p^{-2})$. Since $(K+1)\eta > 1$, $\sum_p |M_p(s) - 1|$ converges uniformly on the compact set, which gives absolute and locally uniform convergence of $\prod_{p \notin S,\, p > p_0} M_p$; the finitely many remaining factors are rational functions of p^{-s} whose only possible poles are at $|p^{-Nw}| = 1$, i.e. on $\Re w = 0$. A uniformly convergent product of non-vanishing holomorphic factors is holomorphic and non-vanishing; hence $M_{f,K}$ vanishes exactly where a local factor does, and $M_p(s) = 0$ iff $F_p(s) = 0$ (the other factors of M_p are $\det(1 - g_p v^N \mid \Psi_N)$, non-zero for $\Re w > 0$). Letting $K \to \infty$ gives the continuation to $\Re s > -1$, the finitely many $L(Nw, \Psi_N)$ being meromorphic by Brauer's theorem. $\square$

In words. D_f is a product of finitely many Artin L-functions evaluated at $N(s+1)$, times a remainder that is harmless in a half-plane that grows with the number of factors extracted but never reaches $\Re s = -1$. Every singularity of D_f to the right of -1 is a zero or pole of an L-function of the Galois group of f, at a predictable place.

Remark 5. $M_{f,K}$ is *not* $1+O(p^{-2})$ at the level of individual primes: the mixed term $2\omega^2 uv/(1+\omega v)$ is of size $p^{-1-\Re w}$. In the numerical work these terms are extracted as well ($[\zeta(w+1)L(w+1,\chi_D)]^4$, $[\zeta(2w+1)L(2w+1,\chi_D)]^{-8}$, … for quadratics) so that the remaining product converges like $\sum p^{-2}$; see `explicit-diag.py`.

2.4 Quadratics: the exponents in closed form

For a quadratic f of discriminant D, $G=\{e,\tau\}$, $\chi_V=\mathbf{1}+\chi_D$, and every virtual character is $a\mathbf{1}+b\chi_D$ with $a,b\in\mathbb{Z}$. Write $\Psi_N=a_N\mathbf{1}+b_N\chi_D$, so that $L(Nw,\Psi_N)=\zeta(Nw)^{a_N}L(Nw,\chi_D)^{b_N}$, and let $M_N(x)=\frac{1}{N}\sum_{j|N}\mu(N/j)x^j$ be the necklace polynomial.

Proposition 6. *For every $N\geq 1$,*

$$a_N+b_N=\Psi_N(e)=-M_N(-2),\qquad a_N-b_N=\Psi_N(\tau)=\frac{1}{N}\sum_{\substack{j|N\\2|j}}\mu(j)(-1)^{N/j+1}2^{N/j}.$$

*In particular $a_N=b_N=\frac{1}{2}M_N(2)$ for odd N, and for even N $a_N+b_N=-M_N(-2)<0$; the first values are those of Table **??**, and $a_1=b_1=1$, i.e. $L(w,\Psi_1)=\zeta_K(w)$; $\Psi_2=-\operatorname{Sym}^2V$; $a_3=b_3=1$, so that $\zeta(3w)L(3w,\chi_D)$ appears in the numerator and D_f has a simple pole at $s=-\frac{2}{3}$; and $a_N>0$ for all odd N, giving a pole of order $a_N=\frac{1}{2}M_N(2)$ at $s=-1+1/N$.*

Proof. $\chi_V(e^j)=2$ and $\chi_V(\tau^j)=2$ or 0 according as j is even or odd; insert into Definition 2, using $(-1)^{k+1}2^k=-(-2)^k$ for the first formula. For odd N there is no even divisor, so $a_N=b_N=-\frac{1}{2}M_N(-2)=\frac{1}{2}M_N(2)$; $M_N(2)>0$ counts binary Lyndon words. □

In words. For a quadratic the L-functions are ζ and $L(s,\chi_D)$ and the exponents are the necklace numbers of Table **??**. Two features matter later: $\zeta(3s+3)$ in the numerator, which gives a real pole at $s=-\frac{2}{3}$ and hence a term $x^{m-2/3}$ in every average of S_f; and $1/(\zeta(2s+2)^2L(2s+2,\chi_D))$, whose poles at half the zeros give the oscillating terms $x^{\rho/2}$.

2.5 The natural boundary

Theorem 7 (natural boundary)**.** *Let f be irreducible of degree $n\geq 1$ without fixed prime divisor. Then D_f has infinitely many zeros in $-1<\Re s<0$, accumulating at every point of the line $\Re s=-1$, and $\Re s=-1$ is a natural boundary of D_f and of $\sum_n A_f(n)n^{-s}=\zeta(s)D_f(s)$. The same holds for $f(t)=t$, i.e. for the generating Dirichlet series of the twin-prime singular series.*

Proof. By (4), for $p\notin S$ with $\omega=\omega_f(p)\geq 1$, $F_p(s)=0$ iff $\omega v=-(1-2\omega u)$, i.e. iff $p^{-w}=-(1-2\omega/p)/\omega$. These are the points

$$s_{p,j}=-1+\frac{\log\omega-\log(1-2\omega/p)}{\log p}+\frac{(2j+1)\pi i}{\log p},\qquad j\in\mathbb{Z},$$

with $\Re s_{p,j}\to -1^+$ as $p\to\infty$ (for $\omega=1$, $\Re s_{p,j}=-1+2/(p\log p)+O(p^{-2})$). By the Chebotarev density theorem there are infinitely many p with $\omega_f(p)\geq 1$ (indeed with $\omega_f(p)=n$). Given $t_0\in\mathbb{R}$ and $\varepsilon>0$, choose such a p with $\log p>\pi/\varepsilon$ and $|\Re s_{p,j}+1|<\varepsilon$; the spacing of the $\Im s_{p,j}$ is $2\pi/\log p<2\varepsilon$, so some $s_{p,j}$ lies within 2ε of $-1+it_0$. By Theorem 4 with K large, in a neighbourhood of $s_{p,j}$ we have $D_f=\Lambda_K\cdot M_{f,K}$ with $\Lambda_K=\prod_{N\leq K}L(Nw,\Psi_N)$ meromorphic and $M_{f,K}(s_{p,j})=0$; so $s_{p,j}$ is a zero of D_f unless it is a pole of Λ_K. The poles of Λ_K form a discrete set, so all but finitely many of the zeros $s_{p,j}$ in any bounded region are zeros of D_f. Now suppose D_f continued meromorphically to a disc B around $-1+it_0$. Zeros of a meromorphic function accumulate only at poles, so $-1+it_0$ would be a pole; but in a punctured neighbourhood of a pole $|D_f|$ is large, whereas zeros $s_{p,j}$ lie arbitrarily close to $-1+it_0$: a contradiction. For $\zeta(s)D_f(s)$ the same argument applies, since $\zeta(s)\neq 0$ on $\Re s=-1$. For $f(t)=t$, $F_p=(1-2u+v)/(1-u)^2$ vanishes at $p^{-w}=-(1-2/p)$ and the argument is identical. □

Remark 8 (no asymptotic explicit formula). Bhowmik and Schlage-Puchta [5] proved that a Dirichlet series with a natural boundary at $\Re s = \sigma_0$ admits no asymptotic expansion of its summatory function by residue terms with error $O(x^{\sigma_0-\varepsilon})$. Applied to $\zeta(s)D_f(s)$ (subject to checking their growth hypotheses in our setting), this says that for every $\varepsilon > 0$ the Riesz mean $R_m(x)$ minus the sum of *all* residues in $\Re s > -1+\varepsilon$ is $\Omega(x^{m-1-\varepsilon})$: the error term $O(x^{m-1+\varepsilon})$ of the twin-prime explicit formula [17] is best possible in this strong sense, and so is ours.

In words. Each local factor vanishes on a lattice of points whose real parts tend to -1 and whose imaginary parts become dense, so D_f cannot be continued across $\Re s = -1$. This is why the explicit formula below is conditional while the twin-prime formula of Goldston and Suriajaya is not: they could move the contour up to the wall without crossing anything dangerous; for non-linear f the L-factors with $N \geq 3$ pile up in front of it.

Remark 9. For $f(t) = t$ the pure-v part of the local factor is $1 + v = (1-v^2)/(1-v)$, so the L-function part of D_f is just $\zeta(w)/\zeta(2w)$ and the contour in [17] can be moved to $\Re s = -1+\varepsilon$, crossing the whole critical strip of $\zeta(2s+2)$: this is why the linear case has an *unconditional* explicit formula with error $O(x^{m-1+\varepsilon})$, and why that error term cannot be improved by contour shifting. For $n \geq 2$ the factors $L(Nw, \Psi_N)$ with $N \geq 3$ pile up between $\Re s = -\frac{3}{4}$ and $\Re s = -1$, and the contour can only be moved a little past the first zero family.

3 The explicit formula for the diagonal

Let $R_m(x) = \sum_{n\leq x}(x-n)^m A_f(n)$, so that for $\Re s > 1$

$$R_m(x) = \frac{1}{2\pi i}\int_{(2)} G_m(s)x^{s+m}\,ds, \qquad G_m(s) = \frac{m!\,\zeta(s)D_f(s)}{s(s+1)\cdots(s+m)}. \tag{6}$$

From now on f is quadratic with discriminant D, so that by Theorem 4 and Proposition 6, for every $K \geq 4$,

$$D_f(s) = \frac{\zeta(w)L(w,\chi_D)\,\zeta(3w)L(3w,\chi_D)}{\zeta(2w)^2L(2w,\chi_D)}\prod_{N=4}^{K}\zeta(Nw)^{a_N}L(Nw,\chi_D)^{b_N}\;M_{f,K}(s), \qquad w = s+1. \tag{7}$$

3.1 Statement

Theorem 10. *Let f be an irreducible quadratic of discriminant D and $m \geq 2$. Assume the Riemann hypothesis for ζ and for $L(s,\chi_D)$. Then there is $\varepsilon_0 \in (0, \frac{1}{100})$ such that for $x \geq 2$ and some $U \in [x^{16}, 2x^{16}]$,*

$$R_m(x) = \frac{D_f(1)}{m+1}x^{m+1} + x^m\Big(-\frac{\log x}{2C(f)} + c_{f,m}\Big) + c'_{f,m}\,x^{m-2/3} + x^{m-1}\sum_{\substack{\rho=\beta+i\gamma\\|\gamma|\leq 2U}} Q_{m,\rho}(\log x)\,x^{\rho/2} + O(x^{m-3/4-\varepsilon_0}), \tag{8}$$

where ρ runs over the zeros of $\zeta(s)^2L(s,\chi_D)$ (all with $\beta = \frac{1}{2}$), $Q_{m,\rho}$ is the polynomial (of degree less than the order of the pole of G_m at $\rho/2-1$, i.e. of degree 0 at simple zeros of L and 1 at simple zeros of ζ) such that $Q_{m,\rho}(\log x)x^{\rho/2-1+m}$ is the residue of $G_m(s)x^{s+m}$ at $s = \rho/2-1$, and

$$c'_{f,m} = \operatorname*{Res}_{s=-2/3} G_m(s) = \frac{m!\,\zeta(-\frac{2}{3})\zeta(\frac{1}{3})L(\frac{1}{3},\chi_D)L(1,\chi_D)}{3\,\zeta(\frac{2}{3})^2L(\frac{2}{3},\chi_D)}\prod_{N\geq 4}\zeta(\tfrac{N}{3})^{a_N}L(\tfrac{N}{3},\chi_D)^{b_N}\frac{M_{f,K}(-\frac{2}{3})}{(-\frac{2}{3})(\frac{1}{3})\cdots(m-\frac{2}{3})},$$

$c_{f,m}$ being the constant term of the residue at $s = 0$. The implied constant depends on f, m and ε_0.

Remark 11 (the class number in the term $x^{m-2/3}$). The coefficient $c'_{f,m}$ contains the factor $L(1,\chi_D)$, which for $D<0$ equals $2\pi h(D)/(w_D\sqrt{|D|})$ by Dirichlet's class number formula: the class number of $\mathbb{Q}(\sqrt{D})$ enters the second moment of the pair singular series of f through the real term $x^{m-2/3}$, alongside the values $\zeta(\frac{1}{3})$, $L(\frac{1}{3},\chi_D)$ and the Euler product $M_{f,K}(-\frac{2}{3})$.

The coefficient of $x^m\log x$ is $-1/(2C(f))$ because $\operatorname{Res}_{s=0} D_f = 1/C(f)$ (part I, Theorem 3) and $\zeta(0)=-\frac{1}{2}$; explicitly $c_{f,m} = -\frac{1}{2C(f)}\big(\log 2\pi - H_m\big) - \frac{1}{2}\kappa_f$ with κ_f the constant term of the Laurent expansion of D_f at 0 and $H_m=\sum_{k\le m} 1/k$.

In words. After the expected terms x^{m+1}, $x^m\log x$, x^m comes an unexpected real term $x^{m-2/3}$, then oscillating terms of size $x^{m-3/4}$ with frequencies $\gamma/2$ in $\log x$, γ the ordinates of the zeros of ζ and $L(s,\chi_D)$; the rest is smaller by a fixed power of x. The hypothesis is needed because the proof evaluates $1/\zeta$ and $1/L$ just left of the critical line.

3.2 Growth lemmas

We use the following consequences of the Riemann hypothesis for ζ and $L(s,\chi_D)$ ([37, Chapter 14] for ζ; the proofs transfer verbatim to $L(s,\chi_D)$, see [24, §5.7]). Let $\Lambda\in\{\zeta, L(\cdot,\chi_D)\}$ and $t\ge 2$.

(G1) For every fixed $\sigma>\frac{1}{2}$: $\Lambda(\sigma+it)^{\pm1}\ll t^{\varepsilon}$; for every fixed σ: $\Lambda(\sigma+it)\ll t^{\max(0,1/2-\sigma)+\varepsilon}$, and for fixed $\sigma<\frac{1}{2}$: $\Lambda(\sigma+it)^{-1}\ll t^{\varepsilon}$ (by the functional equation, $|\Lambda(\sigma+it)|\asymp t^{1/2-\sigma}|\Lambda(1-\sigma-it)|$).

(G2) Unconditionally, for $-1\le\sigma\le 2$: $\dfrac{\Lambda'}{\Lambda}(\sigma+it)=\sum_{|\gamma-t|\le 1}\dfrac{1}{\sigma+it-\rho}+O(\log t)$, the sum over zeros $\rho=\beta+i\gamma$ of Λ, and the number of zeros with $|\gamma-t|\le 1$ is $\le c_0\log t$.

Lemma 12 (horizontal segments)**.** *Assume RH for ζ and $L(s,\chi_D)$. For every $\delta>0$ there is $\varepsilon_1(\delta)>0$ such that for all $T\ge T_0(\delta)$ and $\varepsilon_0\le\varepsilon_1$, there exists $U\in[T,2T]$ with*

$$\big|\zeta(\sigma+2iU)^{-1}\big|+\big|L(\sigma+2iU,\chi_D)^{-1}\big|\le T^{\delta}\qquad \textit{uniformly for } \tfrac{1}{2}-2\varepsilon_0\le\sigma\le 2.$$

Proof. Let Λ be one of the two functions, $\sigma_1=\frac{1}{2}+\varepsilon_0$, and $Z=[\frac{1}{2}-2\varepsilon_0,\sigma_1]$ (the "zone"). For $\sigma\ge\sigma_1$ the claim is (G1) with $\delta/2$ in place of ε. For $\sigma\in Z$ and $t=2U$ write, by (G2),

$$\log\frac{1}{|\Lambda(\sigma+it)|}=\log\frac{1}{|\Lambda(\sigma_1+it)|}+\Re\int_{\sigma}^{\sigma_1}\frac{\Lambda'}{\Lambda}(\sigma'+it)\,d\sigma'\le\frac{\delta}{4}\log T+\sum_{|\gamma-t|\le 1}\int_Z\frac{d\sigma'}{|\sigma'+it-\rho|}+3\varepsilon_0 C\log t.$$

Under RH $\beta=\frac{1}{2}$, so $|\sigma'+it-\rho|=\sqrt{(\sigma'-\frac{1}{2})^2+(t-\gamma)^2}$ and, with $\theta=|t-\gamma|$, $\int_Z\le\int_{-2\varepsilon_0}^{2\varepsilon_0}\frac{dy}{\sqrt{y^2+\theta^2}}=2\operatorname{arsinh}(2\varepsilon_0/\theta)$, which is $\le 4\varepsilon_0/\theta$ for $\theta\ge 2\varepsilon_0$ and $\le 2\log(4\varepsilon_0/\theta)+2$ for $\theta<2\varepsilon_0$. By (G2) the zeros with $2\varepsilon_0\le\theta\le 1$ contribute at most $4\varepsilon_0\sum\theta^{-1}\le 4\varepsilon_0\log(1/2\varepsilon_0)\,c_0\log t$ (split the range of θ dyadically: each dyadic block $[\theta_0,2\theta_0]$ contains at most $c_0\log t$ zeros, each contributing at most θ_0^{-1}). The zeros with $\theta<2\varepsilon_0$ contribute $2\Sigma_\Lambda(t)+2\#\{\gamma:|\gamma-t|<2\varepsilon_0\}$ with $\Sigma_\Lambda(t)=\sum_{|\gamma-t|<2\varepsilon_0}\log(4\varepsilon_0/|t-\gamma|)$. Now average over $t\in[2T,4T]$: each zero contributes at most $\int_{-2\varepsilon_0}^{2\varepsilon_0}\log(4\varepsilon_0/|y|)\,dy=4\varepsilon_0(1+\log 2)$, and there are $\le c_0'T\log T$ zeros with $\gamma\in[2T-1,4T+1]$, so $\frac{1}{2T}\int_{2T}^{4T}\Sigma_\Lambda(t)\,dt\le 6c_0'\varepsilon_0\log T$ and similarly the average of $\#\{\gamma:|\gamma-t|<2\varepsilon_0\}$ is $\le 4c_0'\varepsilon_0\log T$. By Markov's inequality the set of $t\in[2T,4T]$ with $\Sigma_\zeta(t)+\Sigma_L(t)+\#_\zeta+\#_L\le 100c_0'\varepsilon_0\log T$ has measure $\ge\frac{3}{5}\cdot 2T$; pick U with $2U$ in it. Altogether $\log(1/|\Lambda(\sigma+2iU)|)\le(\frac{\delta}{4}+c_1\varepsilon_0\log(1/\varepsilon_0))\log T$ for $\sigma\in Z$, with c_1 absolute, which is $\le\delta\log T$ once $\varepsilon_0\le\varepsilon_1(\delta)$. ☐

3.3 Proof of Theorem 10

Fix $K=15$ (any $K\ge 4$ with $(K+1)(\frac{1}{4}-\varepsilon_0)>1$ works), $\delta=\varepsilon_0$ and $\varepsilon_0\le\min(\varepsilon_1(\varepsilon_0),\frac{1}{100})$, which is possible since $\varepsilon_1(\delta)\gg\delta/\log(1/\delta)$. Let U be as in Lemma 12 with $T=x^{16}$, and let

$\mathcal{R}$ be the rectangle with vertices $2 \pm iU$ and $-\frac{3}{4} - \varepsilon_0 \pm iU$. By (6) (truncating the integral at height U costs $\ll x^{m+2}U^{-m}$, since $G_m(2+it) \ll t^{-m-1}$) and the residue theorem,

$$R_m(x) = \sum_{\text{poles in } \mathcal{R}} \operatorname{Res}\big(G_m(s)x^{s+m}\big) + \frac{1}{2\pi i}\Big(\int_{-3/4-\varepsilon_0-iU}^{-3/4-\varepsilon_0+iU} + \int_{\text{horiz.}}\Big) G_m(s)x^{s+m}\,ds + O(x^{m+2}U^{-m}).$$

Poles. By (7) and Theorem 4, in $\Re s \geq -\frac{3}{4} - \varepsilon_0 > -\frac{4}{5}$ the poles of G_m are: $s = 1$ (simple, from $\zeta(s)$; $D_f(1) = \sum_d a_f(d)/d$), $s = 0$ (double: the pole of $\zeta(w)$ and the factor $1/s$), $s = -\frac{2}{3}$ (simple, from $\zeta(3w)$), and $s = \rho/2 - 1$ for the zeros ρ of $\zeta(2w)^2 L(2w, \chi_D)$ with $|\gamma| \leq 2U$; the factors $\zeta(Nw)^{a_N} L(Nw, \chi_D)^{b_N}$, $N \geq 4$, have $\Re(Nw) \geq 1 - 4\varepsilon_0 > \frac{1}{2}$ there, so under RH they have neither zeros nor poles except $\zeta(Nw)$ at $Nw = 1$, i.e. $s = -1 + 1/N \leq -\frac{3}{4}$, outside $\mathcal{R}$ for $N \geq 4$ (and $N = 4$ gives a zero of D_f); $M_{f,K}$ is holomorphic in $\Re s > -1 + \frac{1}{16}$; and $s = -1, \ldots, -m$ are outside $\mathcal{R}$. The residues at 1, 0, $-\frac{2}{3}$ are the three explicit terms of (8), and the residue at $\rho/2 - 1$ is $Q_{m,\rho}(\log x)x^{\rho/2-1+m}$ by definition.

Vertical line. On $\Re s = -\frac{3}{4} - \varepsilon_0$ (so $\Re w = \frac{1}{4} - \varepsilon_0$, $\Re 2w = \frac{1}{2} - 2\varepsilon_0$, $\Re 3w = \frac{3}{4} - 3\varepsilon_0$, $\Re Nw \geq 1 - 4\varepsilon_0$ for $N \geq 4$), (G1) gives, for $|t| \geq 2$, $\zeta(s) \ll |t|^{5/4+\varepsilon_0+\varepsilon}$, $\zeta(w)L(w, \chi_D) \ll |t|^{1/2+2\varepsilon_0+\varepsilon}$, $\zeta(2w)^{-2}L(2w, \chi_D)^{-1} \ll |t|^{\varepsilon}$, $\zeta(3w)L(3w, \chi_D) \ll |t|^{\varepsilon}$, $\prod_{N\geq 4} \ll |t|^{\varepsilon}$, and $M_{f,K} \ll 1$; hence $G_m(s) \ll |t|^{7/4+3\varepsilon_0+\varepsilon-(m+1)}$, and for $m \geq 2$ the exponent is < -1. The vertical integral is therefore $\ll x^{m-3/4-\varepsilon_0}$.

Horizontal segments. On $\Im s = \pm U$, $-\frac{3}{4} - \varepsilon_0 \leq \sigma \leq 2$: (G1) bounds $\zeta(s)$, $\zeta(w)L(w)$, $\zeta(3w)L(3w)$ and the $N \geq 4$ factors by $U^{7/4+3\varepsilon_0+\varepsilon}$ (worst case at the left end), Lemma 12 bounds $\zeta(2w)^{-2}L(2w, \chi_D)^{-1}$ by $U^{3\varepsilon_0}$, and $M_{f,K} \ll 1$; the kernel is $\ll U^{-m-1}$ and $|x^{s+m}| \leq x^{m+2}$. The two segments contribute $\ll x^{m+2}U^{7/4+6\varepsilon_0+\varepsilon-m-1} \leq x^{m+2}U^{-1/4+7\varepsilon_0}$ for $m \geq 2$, and the truncation of the original integral at height U contributes $\ll x^{m+2}U^{-m}$; with $U \geq x^{16}$ and $\varepsilon_0 \leq \frac{1}{100}$ both are $\ll x^{m-1}$.

Collecting, $R_m(x)$ equals the residue sum plus $O(x^{m-3/4-\varepsilon_0})$, which is (8). $\square$

Remark 13 (on the zero sum). The sum over $|\gamma| \leq 2U$ is not absolutely convergent as $U \to \infty$ with the polynomial growth of the coefficients; as in [17, Theorem 3], simplicity of the zeros and a bound $\sum_{|\gamma|\leq T} |\zeta'(\rho)|^{-2} \ll T$ (Gonek–Hejhal) would make $x^{m-3/4}\sum_\gamma Q_{m,\rho}x^{i\gamma/2}$ absolutely convergent for $m \geq 2$. Numerically the coefficients decay like $|\gamma|^{-5/4}$ for $m = 2$.

3.4 General f

Theorem 14 (general f)**.** *Let f be irreducible of degree $n \geq 2$ and $m \geq 2$. Assume that the Artin L-functions of the irreducible constituents of $\Psi_1 = \chi_V$, $\Psi_2 = -\chi_{\mathrm{Sym}^2 V}$ and $\Psi_3 = \frac{1}{3}(\chi_V^3 - \psi^3\chi_V)$ are entire (automatic when G is an M-group, e.g. S_3, S_4, A_4, D_n) and satisfy the Riemann hypothesis. Then there is $\varepsilon_0 > 0$ such that, for $U \in [x^{16}, 2x^{16}]$ suitably chosen,*

$$R_m(x) = \frac{D_f(1)}{m+1}x^{m+1} + x^m\Big(-\frac{\log x}{2C(f)} + c_{f,m}\Big) + c'_{f,m}\,x^{m-2/3} + x^{m-1}\sum_{\substack{\rho \\ |\gamma|\leq 2U}} Q_{m,\rho}(\log x)\,x^{\rho/2} + O(x^{m-3/4-\varepsilon_0}),$$

where ρ runs over the zeros of $L(s, \mathrm{Sym}^2 V) = \zeta_K(s)\prod_O \zeta_{K_O}(s)$, and $c'_{f,m} = 0$ unless the trivial character occurs in Ψ_3, in which case $c'_{f,m}x^{m-2/3}$ is the residue at $s = -\frac{2}{3}$. For $G = S_n$ the trivial character occurs in Ψ_3 with multiplicity 1, so the term $x^{m-2/3}$ is present for every S_n-polynomial.

Proof. Identical to the proof of Theorem 10, with $\zeta(w)L(w, \chi_D)$ replaced by $L(w, \Psi_1) = \zeta_K(w)$, $\zeta(2w)^{-2}L(2w, \chi_D)^{-1}$ by $L(2w, \Psi_2) = 1/L(2w, \mathrm{Sym}^2 V)$, and $\zeta(3w)L(3w, \chi_D)$ by $L(3w, \Psi_3)$: (G1) and (G2) hold for entire Artin L-functions satisfying the Riemann hypothesis with the same proofs (Hadamard product and functional equation), the factors $L(Nw, \Psi_N)$, $N \geq 4$, are evaluated at $\Re(Nw) \geq 1 - 4\varepsilon_0$ where they are $\ll |t|^{\varepsilon}$ and free of zeros and poles, and $M_{f,K}$ is bounded by Theorem 4. The multiplicity of the trivial character in Ψ_3 is $\frac{1}{3}(\langle \chi_V^3, 1\rangle - \langle \psi^3\chi_V, 1\rangle)$; for S_n, $n \geq 3$, $\langle \chi_V^3, 1\rangle = 5$ (orbits of S_n on maps $\{1,2,3\} \to \{1,\ldots,n\}$) and $\langle \psi^3\chi_V, 1\rangle$ is the mean number of fixed points of g^3, i.e. $\mathbb{E}[c_1 + 3c_3] = 1 + 3\cdot\frac{1}{3} = 2$ with c_k the number of k-cycles; for $n = 2$ the values are 4 and 1. □

In words. In any degree the same formula holds with the zeros of the Dedekind zeta functions of the root field and of the pair fields, under the corresponding Riemann hypotheses; the real term $x^{m-2/3}$ is universal for the generic Galois group.

3.5 $f = t^2 + 1$: every coefficient

For $f = t^2 + 1$, $D = -4$, $\chi_D = \chi_{-4}$, $K = \mathbb{Q}(i)$, and every quantity in (8) is computable: $D_f(1) = 0.7507908137$, $C(f) = 1.372813462821$ (from $L(1,\chi_{-4}) = \pi/4$ and $M_{f,K}(0)$; part I quotes $1.3728134628\ldots$), $c_{f,2} = -0.454974899864$, $c_{f,3} = -0.333569719058$, and the coefficients $Q_{m,\rho}$ for the zeros with $\gamma \leq 100$ are computed as Taylor coefficients of $(s-\rho/2+1)^{\mathrm{ord}}G_m(s)$ by trapezoidal Cauchy integrals on circles of radius 0.015 (`explicit-diag.py`; $M_{f,K}$ with $K = 15$ and primes up to $4\cdot 10^6$, with the tails of the mixed terms handled by the prime number theorem). The table of the first coefficients is item (C1) of Appendix A.

3.6 Numerical verification

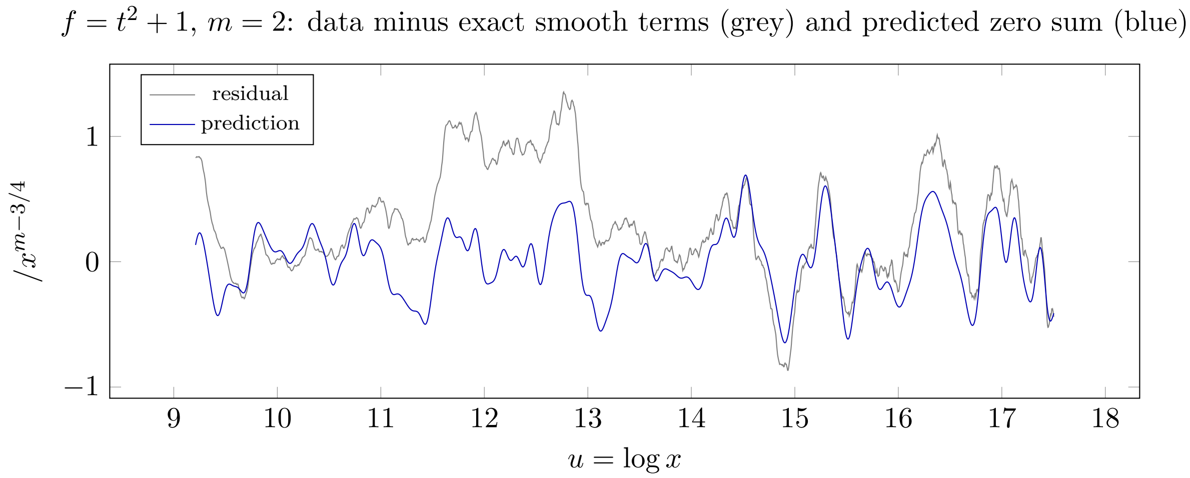

Figure 1: The residual of the diagonal Riesz mean of order 2 of t^2+1 after subtracting the exact terms at $s = 1$, $s = 0$ and the real poles $s = -2/3, -4/5, -6/7$, against the sum over the zeros of $L(s,\chi_{-4})$ and of ζ with $\gamma \leq 100$. Regression coefficient $\beta = 1.008$ (0.991 with a smooth nuisance basis); per family $\beta_\zeta = 1.006$, $\beta_L = 1.020$.

The raw Riesz means $R_m(x)$, $m = 1, 2, 3$, were computed exactly (compensated summation, $A_f(n)$ from $\omega_f(p) = 1 + \chi_{-4}(p)$) at 4096 points equally spaced in $u = \log x \in [\log 10^4, \log 4\cdot 10^7]$. From R_2 we subtract the exact terms at $s = 1$, $s = 0$, $-\frac{2}{3}$, $-\frac{4}{5}$ (order 3) and $-\frac{6}{7}$ (order 9); the only fitted quantity is a correction to the x^{m+1} coefficient, which comes out at $-1.0\cdot 10^{-12}$ relative to $D_f(1)/(m+1)$ and is itself a check of $D_f(1)$ to twelve digits. The residual, divided by $x^{5/4}$, has root mean square 0.528. The predicted zero sum (zeros of $L(s,\chi_{-4})$ and of ζ with $\gamma \leq 100$) has root mean square 0.27. Regressing the residual on the prediction gives $\beta = 1.008$ (Figure 1); with a smooth nuisance basis $x^{m-8/9}(\log x)^j$, $j \leq 3$, absorbing the dropped real poles, $\beta = 0.991$ with $R^2 = 0.55$ and a remaining root mean square of 0.291; the two families separately give $\beta_\zeta = 1.006$ and $\beta_L = 1.020$. Controls in which all ordinates are shifted by $\delta \in [-0.6, 0.6]$ give β between -0.94 and 0.68 and negative R^2. For $m = 3$ the zero terms (rms 0.11) lie below the smooth leftover (rms 0.6); with the nuisance basis $\beta = 0.998$. The remaining rms 0.29 is accounted for by the truncation at $\gamma \le 100$ (the coefficients decay only like $\gamma^{-5/4}$) and the $O(x^{m-3/4-\varepsilon_0})$ term. We regard this as a confirmation of Theorem 10 with the predicted amplitudes and phases to about 1%; it does *not* resolve individual ordinates closer than $4\pi/8.3 \approx 1.5$. When the contour is pushed to $\Re s = -\frac{7}{8}$ and the second family of zero terms is added, the predicted sum has root mean square 13 (zeros of ζ) and 92 (zeros of L) in the same units and regression coefficient 0.001 against the data: the second family is real as a set of residues but cancels against the line integral, as explained after Theorem 7.

4 Ω-results and conditional bounds

Theorem 15. *Let f be an irreducible quadratic and $m \ge 1$, and let $E_m(x) = R_m(x) - \frac{D_f(1)}{m+1}x^{m+1} - x^m\left(-\frac{\log x}{2C(f)} + c_{f,m}\right) - c'_{f,m}x^{m-2/3}$. Suppose that for some zero $\rho_1 = \frac{1}{2} + i\gamma_1$ of $\zeta(s)^2 L(s,\chi_D)$ the function G_m has a genuine pole at $\rho_1/2 - 1$ (i.e. $Q_{m,\rho_1} \ne 0$). Then, unconditionally,*

$$E_m(x) = \Omega_\pm\left(x^{m-3/4}\right).$$

For $f = t^2 + 1$ the hypothesis holds with $\gamma_1 = 6.0209\ldots$, the first zero of $L(s,\chi_{-4})$ ($|Q_{2,\rho_1}|$ is computed in Section 3.5 and is non-zero).

Proof. Suppose $E_m(x) \le \varepsilon x^{m-3/4}$ for all $x \ge x_0$. The function $F(x) = \varepsilon x^{m-3/4} - E_m(x)$ is then non-negative for $x \ge x_0$, and for $\Re s > 1$

$$\int_{x_0}^{\infty} F(x)\, x^{-s-m-1}\, dx = \frac{\varepsilon x_0^{-s-1/4}}{s+\frac{1}{4}} - G_m(s) + \Big(\text{the Mellin transforms of the subtracted terms}\Big) + \big(\text{entire}\big),$$

where we used $\int_1^\infty R_m(x)x^{-s-m-1}dx = G_m(s)$ for $\Re s > 1$ (Mellin inversion of (6)) and the fact that $\int_1^{x_0}$ is entire; the Mellin transforms of x^{m+1}, $x^m \log x$, x^m, $x^{m-2/3}$ are $1/(s-1)$, $1/s^2$, $1/s$, $1/(s+\frac{2}{3})$ up to constants, and they cancel the poles of G_m at 1, 0, $-\frac{2}{3}$ exactly by the definition of the coefficients. The right-hand side is therefore meromorphic in $\Re s > -\frac{4}{5}$ (Theorem 4), holomorphic on the real segment $(-\frac{4}{5}, \infty)$ (the remaining real singularities of D_f in this range are the zeros at $-\frac{1}{2}$ and $-\frac{3}{4}$), and has a pole at $\rho_1/2 - 1$, of real part $-\frac{3}{4}$. By Landau's theorem for integrals with non-negative integrand [31, Theorem 1.7], the abscissa of convergence of the left-hand side is a singularity of the function on the real axis; hence the integral converges for $\Re s > -\frac{4}{5}$, and its value is holomorphic there, contradicting the pole at $\rho_1/2 - 1$. Thus $E_m(x) > \varepsilon x^{m-3/4}$ for arbitrarily large x, i.e. $E_m = \Omega_+(x^{m-3/4})$; applying the argument to $-E_m$ gives Ω_-. □

Remark 16. (i) No positivity of A_f is needed, and no hypothesis beyond the existence of one uncancelled pole on $\Re s = -\frac{3}{4}$; if $\zeta^2 L(\chi_D)$ had a zero with $\beta > \frac{1}{2}$ the same argument gives $\Omega_\pm(x^{m-1+\beta/2})$, provided that pole is uncancelled. (ii) Under RH for ζ and $L(s,\chi_D)$, Theorem 10 gives $E_m(x) \ll x^{m-3/4+\varepsilon}$ for $m \ge 2$ provided the zero sum is $O(x^\varepsilon)$ in the normalisation $x^{m-3/4}$, which follows from the simplicity and Gonek–Hejhal hypotheses of the remark above; so under those hypotheses the exponent $m - \frac{3}{4}$ is exact. (iii) For $f = t^2 + t + 41$ the first zero of $L(s,\chi_{-163})$ has $\gamma_1 = 0.2029$, so the Ω-oscillation has period $4\pi/\gamma_1 \approx 62$ in $\log x$: over any computable range the term looks like a constant multiple of $x^{m-3/4}$. The family $\{L(s,\chi_D)\}$ has symplectic symmetry [25], so such a low zero is the class-number-one exception, not the rule.

In words. Independently of any hypothesis, one uncancelled pole on $\Re s = -\frac{3}{4}$ forces the error term to be as large as $x^{m-3/4}$ in both directions infinitely often: the zeros of $L(s,\chi_D)$ are visible in the second moment of S_f unconditionally.

5 The off-diagonal: what part III must prove

This section turns Hypothesis (E) of part I into the analytic continuation of one Dirichlet series and identifies that series for quadratics.

5.1 The analytic reformulation

For d squarefree and roots $s \neq s'$ of f modulo d let $m_{ss'} \in [1, d-1]$ be the representative of $s' - s$. Define, for $\Re s > 1$,

$$O_f(s) = \sum_d W_f(d) \sum_{\substack{s,s' \bmod d \\ s \neq s'}} \Big(d^{-s}\zeta\Big(s, \frac{m_{ss'}}{d}\Big) - \frac{\zeta(s)}{d}\Big), \tag{9}$$

$\zeta(s,\alpha)$ the Hurwitz zeta function, so that $d^{-s}\zeta(s, m/d) = \sum_{h \equiv m\ (d)} h^{-s}$; each bracket is entire in s, the poles at $s = 1$ cancelling.

Proposition 17. *(i) For $\Re s > 1$,*

$$\sum_{h \geq 1} \big(S_f(h) - C(f)^2\big) h^{-s} = C(f)^2\, \zeta(s) \big(D_f(s) - D_f(1)\big) + C(f)^2\, O_f(s),$$

the left side converging absolutely and the double series (9) converging in the indicated order. (ii) If O_f continues holomorphically to $\Re s > -\delta$ for some $\delta > 0$ with $O_f(\sigma + it) \ll (1 + |t|)^A$ there, then for some $\delta' > 0$

$$\sum_{h \leq H} \Big(1 - \frac{h}{H}\Big)\big(S_f(h) - C^2\big) = -\tfrac{1}{2} C(f) \log H + A_f^* + O(H^{-\delta'}), \qquad A_f^* = C^2\big(\kappa_f^* + O_f(0)\big),$$

κ_f^* the explicit constant produced by the diagonal. In particular Conjecture (3) holds in Cesàro form and Hypothesis (E) in Cesàro form, $\mathrm{Off}_f^*(H) = O(H)$, holds.

Proof. (i) By part I, Theorem 2, $S_f(h)/C^2 = \sum_q b(q) c_q^f(h)$ absolutely, and the sum over h in a residue class regroups, for each squarefree d, into pairs of roots modulo d, the subtraction of the mean per pair being allowed because $\sum_{d \mid q} \mu(q/d) = 0$ for $q > 1$. With h^{-s} in place of the indicator of $h \leq H$, $\sum_{h \equiv m (d)} h^{-s} = d^{-s}\zeta(s, m/d)$ for $m \in [1, d-1]$ and $d^{-s}\zeta(s)$ for $m \equiv 0$, and the subtracted mean becomes $\zeta(s)/d$. The diagonal pairs give $\sum_d a_f(d) \zeta(s)(d^{-s} - d^{-1}) = \zeta(s)(D_f(s) - D_f(1))$ and the remaining pairs give $O_f(s)$. Absolute convergence of the left side for $\Re s > 1$ follows from $\sum_{h \leq H} S_f(h) = C^2 H + O(\log H)$ (part I, Theorem 2) by partial summation.
(ii) By Perron's formula with the weight $1 - h/H$,

$$\sum_{h \leq H} \Big(1 - \frac{h}{H}\Big)\big(S_f(h) - C^2\big) = \frac{C^2}{2\pi i} \int_{(2)} \big[\zeta(s)(D_f(s) - D_f(1)) + O_f(s)\big] \frac{H^s}{s(s+1)}\, ds.$$

Move the contour to $\Re s = -\delta'$, $0 < \delta' < \min(\delta, \frac{1}{2})$. The diagonal factor $\zeta(s)(D_f(s) - D_f(1))$ is holomorphic there except at $s = 0$ (a simple pole of D_f times the kernel's $1/s$; $D_f(s) - D_f(1)$ vanishes at $s = 1$, so there is no pole at 1), where the residue is $-\frac{1}{2C} \log H + \kappa_f^*$ because $\zeta(0) = -\frac{1}{2}$ and $\mathrm{Res}_0\, D_f = 1/C$; the off-diagonal factor is holomorphic and contributes the residue $O_f(0)$ at the pole of the kernel. On $\Re s = -\delta'$ the diagonal factor has polynomial growth ($D_f = \zeta_K(w) \times$ an absolutely convergent product for $\Re s > -\frac{1}{2}$, Theorem 4 with $K = 1$), and so has O_f by hypothesis; the kernel makes the integral $O(H^{-\delta'})$. $\square$

In words. Conjecture (3) is the statement that one Dirichlet series, O_f, has no singularity at $s = 0$ and none slightly to its left. A pole of O_f at $s = 0$ would add a term $H \log H$ to the Cesàro sum and break the universality of $-\frac{1}{2}$; this is the danger for Galois groups that are not 2-transitive noted in part I, and (9) is where it would show.

5.2 Quadratics: a Dirichlet series of Salié-type sums

Let $f = at^2 + bt + c$ with discriminant D. For d coprime to $2aD$ the roots of f modulo d correspond to the square roots of $D' = D/a^2$ modulo d, and the differences of distinct roots modulo d are indexed by divisors $d' \mid d$, $d' > 1$, and square roots ν of D' modulo d' (part I, companion note, Proposition A1). Hurwitz's formula, valid for $\Re s < 0$,

$$\zeta(s, \alpha) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sum_{k \geq 1} \frac{\sin(2\pi k\alpha + \pi s/2)}{k^{1-s}},$$

turns each term $d^{-s}\zeta(s, m/d)$ into a Fourier series in m/d, and hence $O_f(s)$, *if* it continues to $\Re s < 0$, into

$$O_f(s) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sum_{k \geq 1} \frac{1}{k^{1-s}} \sum_{d} \frac{W_f(d)}{d^s} \sum_{\substack{d' \mid d \\ d' > 1}} 2^{k(d)-k(d')} \sum_{\nu^2 \equiv D' \ (d')} \sin\Big(2\pi k \, \frac{\overline{(d/d')}\,\nu}{d'} + \frac{\pi s}{2}\Big), \quad (10)$$

$k(\cdot)$ the number of prime factors and $\overline{(d/d')}$ the inverse of d/d' modulo d'. The innermost sum is a Weyl sum of quadratic roots, a Salié-type sum: for prime d' it is $\sum_{\nu^2 \equiv D'} e(k\nu/d')$ up to the phase. Dirichlet series in d of such sums, for fixed k and without the weights W_f, are the objects of Hooley [21], Bykovskii, Duke, Friedlander and Iwaniec [11, 12] and Tóth [38]: their continuation is the equidistribution of the roots of $\nu^2 \equiv D'$ modulo d, and its spectral source is the theory of Maass forms through the Kuznetsov formula and its Salié-sum analogue. The identity (10) is formal (it interchanges the sums over d and k); making it rigorous is the content of the following programme.

Proposition 18 (the two estimates)**.** *Hypothesis (E) in Cesàro form holds for the quadratic f if the following two statements hold.*

(E1) (Small moduli.) For every $k \geq 1$ and every unit class, the weighted Weyl sums $\sum_{d \leq x} \lambda(d) \sum_{\nu^2 \equiv D'(d)} e(k\bar{u}\nu/$ with λ the multiplicative weight $\lambda(p) = p/(p-4)$ on split primes (so that $|W_f(d)| \asymp \lambda(d)/d$), are $\ll x(\log x)^{-1-\eta}$ uniformly in $k \leq x^c$, for some $\eta, c > 0$.

(E2) (Large moduli.) $\displaystyle\sum_{h \leq H} \sum_{\substack{e \mid Q(h) \\ e < Q(h)/H}} \frac{e\,\lambda(Q(h)/e)}{Q(h)} - H \sum_{d > H} \frac{W_f(d)\,\omega_f^\circ(d)}{d} = O(1)$, *where $Q(h) = a^2h^2 - D$ and $\omega_f^\circ = \omega_f^2 - \omega_f$.*

Proof. This is Propositions A1–A3 of the companion note to part I: the moduli $d \leq H$ contribute a weighted sum of discrepancies of the roots at the points $\{H/d\}$, which (E1) controls through the Erdős–Turán inequality, and the moduli $d > H$ contribute the hyperbola sum of (E2), because a pair of distinct roots modulo $d > H$ has a difference $\leq H$ only if every prime factor of d divides $hQ(h)$ for some $h \leq H$. □

5.3 What Hooley already did, and the route

The decomposition (2) is not new in spirit: Hooley's 1963 asymptotic for $\sum_{n \leq x} \tau(n^2 + a)$ [21] splits the count, after the hyperbola reflection at $\sqrt{n^2 + a}$, into a Dirichlet-series part $\sum_{k \leq X} \rho(k)/k$ (our diagonal at $s = 1$) and a sum $\sum_k \{2\Psi_k(x) - \Psi_k(Y_k)\}$ of sawtooth values $\Psi_k(y) = \sum_{\nu^2 \equiv -a\,(k)} \psi((y - \nu)/k)$ at the roots (our Off_f, with Y_k the reflected argument giving our large-moduli part). He notes that his untwisted Weyl-sum bound does not apply to these sums because of the factors depending on k, and proves a twisted bound (his Lemma 5) for $\sum_{k \leq x} \rho(h, k) e(\pm hx/k)$ by building the phase into the parametrisation of the roots by binary quadratic forms and into the resulting incomplete Kloosterman sums. That twisted sum, with the phase $e(kH/d)$ and our weights, is what (E1) needs. The route to (E1) and (E2) is therefore:

(A) a bound $\sum_{d\le X,\ e|d}\rho(k,d)\,e(kH/d)\ll k^Ae^BX^{1-\delta}$ for moduli in a progression, with polynomial dependence on k and e: for $d\asymp C\ge(kH)^{1/2+\varepsilon}$ the phase is smooth at scale C and the theorem of Duke, Friedlander and Iwaniec [12] (smooth weights, moduli $\equiv 0 \bmod q$, either sign of D) applies as it stands; for smaller C one must redo Hooley's Lemma 5 with the phase, or treat $e(kH\cdot\Im\gamma i)$ as part of the test function of the Poincaré series of Bykovskii [8]; Iwaniec [23] gives the case $e\mid d$ for $D=-1$ with the uniformity needed for a sieve;

(B) the weights: $W_f(d)=d^{-1}\sum_{e|d}g(e)$ with $g(e)\ll e^{-1+\varepsilon}\tau(e)^A$, so that only $e\le(\log H)^A$ matter and polynomial dependence on e in (A) suffices; no theorem with genuinely multiplicative weights on composite moduli is needed, which is fortunate because none exists;

(C) the large moduli: Hooley's reflection turns them into the same structure with the divisors e of $Q(h)$ in place of d, and the level-of-distribution input for $\#\{h\le H : e\mid Q(h)\}$ is in [23] and, uniformly in the shift, in the recent work of Grimmelt and Merikoski.

Expanding the sawtooth to length K (Vaaler) and using (A)–(C) gives $\mathrm{Off}_f(H)\ll K^A+(\log H)/K$, i.e. $\mathrm{Off}_f(H)=o(\log H)$: this is Hypothesis (E) in Cesàro form, hence Conjecture (3) in Cesàro form for every quadratic, since S_2 is 2-transitive on the two roots. The sharp form $\mathrm{Off}_f(H)=O(1)$ needs decay in the frequency k that the $k^{1/4}$ of [12] does not provide; it is the spectral problem of the next remark.

Remark 19 (a second spectrum). The weighted counts of roots of quadratic congruences summed over moduli have an explicit formula over the Laplace eigenvalues of the modular surface: Bykovskii's formula, in the form of Soundararajan and Young [36, Prop. 2.2], writes the prime geodesic counting function as $2\sum_{n\le X}\sqrt{n^2-4}\,L(1,n^2-4)$ with $L(s,\delta)=\frac{\zeta(2s)}{\zeta(s)}\sum_q\rho_q(\delta)q^{-s}$, $\rho_q(\delta)$ the number of roots of $x^2\equiv\delta$ modulo $4q$, and its explicit formula runs over the Maass eigenvalues; Hejhal [20] treats the Weyl sums for $D>0$ as integrals of automorphic functions over closed geodesics. In our setting the discriminant is fixed and the modulus varies, the regime of Poincaré series and long horocycles (Bykovskii; Marklof and Welsh [29]); $\mathrm{Off}_f(H)$ is a Poincaré series with a y-dependent sawtooth test function, and its natural expansion is $\sum_j c_j(D)\hat{h}_H(t_j)+(\text{Eisenstein})$, the Eisenstein term reintroducing the zeros of ζ. The dichotomy "diagonal governed by zeros of ζ, off-diagonal by the discrete spectrum" is the one of the additive divisor problem and of Motohashi's formula for the fourth moment [33]. Whether the discrete spectrum really contributes to Off_f, or decouples as in the non-dihedral cases of Templier and Tsimerman, is testable: for $f=t^2+1$ the Fourier transform of $\mathrm{Off}_f(e^u)$ in u should, or should not, show lines at the Laplace parameters $t_j=9.5337, 12.1730, 13.7798,\dots$ of $\mathrm{SL}_2(\mathbb{Z})$ beside the lines $\gamma/2$ of the zeros of ζ (item (C4) of Appendix A).

In words. For quadratics the off-diagonal is a question about square roots modulo composite numbers: do the square roots of D modulo d, weighted by a divisor-like function of d, spread out evenly? Hooley proved this without weights, Duke, Friedlander and Iwaniec much more for prime moduli; the weights and the fractional-part test function are the new features, and part III will prove (E1) and (E2). For Galois groups with three or more orbits on pairs of roots, the analogue of (10) has several independent families of Weyl sums, and a bias between them could produce a pole of O_f at $s=0$; we do not know whether this happens.

6 What carries over to $\mathbb{F}_q[u]$

Let $\mathcal{A}=\mathbb{F}_q[u]$, q odd, $f\in\mathcal{A}[t]$ irreducible and separable in t, and define ω_f, P_P, W_f, a_f, D_f as over $\mathbb{Z}$ with $|P|=q^{\deg P}$ in place of p; put $T=q^{-s}$.

Theorem 20. *Theorem 4 holds verbatim over $\mathcal{A}$: $D_f(s)=\prod_{N\le K}L(N(s+1),\Psi_N)\,M_{f,K}(s)$ with the same virtual characters Ψ_N of the Galois group of f over $\mathbb{F}_q(u)$, the L-functions being*

Artin L-functions over $\mathbb{F}_q(u)$ (rational functions of T^N by Weil), and $M_{f,K}$ an Euler product over the primes of $\mathcal{A}$ converging absolutely for $|T| < q^{1-1/(K+1)}$. The circle $|T| = q$ is a natural boundary; in particular D_f is not a rational function of T when $\deg_t f \geq 2$.

Proof. The local computation (4) and Lemma 3 are identical with $u = |P|^{-1}$, $v = |P|^{-(s+1)}$; Chebotarev over $\mathbb{F}_q(u)$ supplies infinitely many primes with $\omega_f(P) \geq 1$, and the zeros of F_P have $|T| \to q$ with arguments $(2j+1)\pi/\deg P$, dense as $\deg P \to \infty$. ☐

In words. The function field does not make the generating function rational, as one might hope; the structure and the wall are the same. What it gives is an *unconditional* explicit formula: the analogue of $R_m(x)$ is a coefficient extraction, a contour integral over a circle $|T| = r$, and Weil's theorem places every zero of every $L(T^N, \Psi_N)$ on an explicit circle, so the residue computation of Section 3 goes through with no hypothesis. Writing it out for $f = t^2 - D$ and comparing with the exact data of part I is item (C5) of Appendix A. (The companion note to part I claimed that the remaining Euler product converges for $|T| < q$; this is wrong, for the same reason as over $\mathbb{Z}$.)

7 Consequences for prime values, conditionally

Everything in this section assumes the Hardy–Littlewood pair conjecture for f with a power-saving error term. Let $\mathbf{1}_t$ be the indicator that $f(t)$ is prime, $A_T = \sum_{t<T} \mathbf{1}_t$, and suppose, as in Montgomery–Soundararajan [30], that $\sum_{t<T} \mathbf{1}_t \mathbf{1}_{t+h} = S_f(h) \sum_{t<T-h} w_t w_{t+h} + O(T^{1/2+\varepsilon})$ uniformly for $h \leq T$, $w_t = 1/\log f(t)$. Then formula (6) of part I expresses $\mathrm{Var}(A_T)$ through $\sum_{h<T}(S_f(h) - C^2)W_T(h)$ with a smooth weight W_T, and inserting the explicit formula gives, provided Conjecture (3) holds with the constant A_f,

$$\frac{\mathrm{Var}(A_T)}{\mathbb{E}(A_T)} = 1 - \frac{\log T}{\log X} + \frac{1 + 2A_f/C(f)}{\log X} + \frac{c_1(f)\, T^{-2/3} + \sum_\rho c_\rho(f,T)\, T^{-3/4}}{\log X} + O\Big(\frac{T^{-1/2+\varepsilon}}{\log X}\Big), \qquad X \approx f(T),$$

with $c_\rho(f,T)$ bounded and oscillating in $\log T$. Three remarks. (i) The first three terms are the polynomial Montgomery–Soundararajan law of part I; A_f consists of an explicit diagonal part and $C^2 O_f(0)$ (Proposition 17). (ii) The zero-induced terms have relative size $T^{-3/4}$, below the $T^{-1/2+\varepsilon}$ that the hypothesis tolerates: nothing about primes and zeros follows, and we do not claim it. (iii) Under GRH the variance side is related to pair correlation of zeros by the Goldston–Montgomery theorem; for $\zeta_K = \zeta L(\chi_D)$ the Selberg-class version [7] applies to each primitive factor, and Murty and Perelli [34] predict that the zeros of the two factors are uncorrelated; but this concerns prime *ideals* of K, for which a Dirichlet series exists, not prime values of f, for which none does.

8 Discussion and open problems

What we learned. (1) The object behind the second moment of prime values of f is not one L-function but the plethystic expansion of $1 + \chi_V v$: a canonical sequence of virtual characters Ψ_N of the Galois group, with $\Psi_1 = V$ and $\Psi_2 = -\operatorname{Sym}^2 V$. (2) The linear case is degenerate: its expansion terminates, which is why the twin-prime explicit formula is unconditional and why the wall at $\Re s = -1$ was never in the way. (3) For non-linear f there is a term $x^{m-2/3}$ larger than the zero terms; a numerical search for the zeros that does not subtract it fails, which is what happened in part I. (4) Near a natural boundary an explicit formula is asymptotic, not convergent, in the position of the contour. (5) With the smooth terms subtracted exactly, the zeros of $L(s, \chi_{-4})$ appear in the second moment of $t^2 + 1$ with the predicted amplitude and phase.

Open problems. (a) Prove (E1) and (E2) for quadratics (part III). (b) The multiplicity of the trivial character in Ψ_N for general G, hence the full list of real poles $-1 + 1/N$; for S_n it is $\frac{1}{N}\sum_{j|N}\mu(j)(-1)^{N/j+1}\,\mathbb{E}[\chi_V(g^j)^{N/j}]$, with moments of fixed-point counts of powers of random permutations. (c) The limiting logarithmic distribution of $E_m(x)/x^{m-3/4}$ (in the sense of Rubinstein–Sarnak) and whether it is biased. (d) Higher moments: the k-point singular series of Montgomery–Soundararajan for polynomial tuples and the virtual characters that appear. (e) The unconditional function-field explicit formula for $t^2 - D$ against the exact data. (f) Whether Hypothesis (E) can fail for non-2-transitive Galois groups: each orbit O of ordered pairs gives a root sum of the difference resolvent $R_O(y) = \prod_O(y-(s'-s))$, quadratic for quadratics, reducible with quadratic factors for V_4 and D_4 quartics ($x^4 + 1$: $R_O = (y^2 - 2)(y^2 + 2)$), irreducible of degree 4 for the cyclic quartic, where only qualitative equidistribution is known (Hooley [22], Kowalski–Soundararajan [26]); the test is whether $\Sigma_f(H) + \frac{1}{2}C\log H$ converges for $x^4 + 1$, $x^4 - 2$, $x^4 + x^3 + x^2 + x + 1$ and a generic S_4 quartic. (g) By the general theorem of Akbary, Ng and Shahabi [1], the normalised error $E_m(x)/(x^{m-3/4}\log x)$ has a limiting logarithmic distribution under RH and a Gonek–Hejhal-type bound; under linear independence of the ordinates of $\zeta\cdot\zeta_{\mathbb{Q}(\sqrt{D})}$ it is symmetric, so E_m is negative with logarithmic density exactly $\frac{1}{2}$ despite the negative main term of (3): a "no bias" statement worth proving. (h) A general natural-boundary criterion for two-variable Frobenian Euler products $\prod_p W(\mathrm{Frob}_p; p^{-1}, p^{-s})$, extending Kurokawa–Moroz; we have proved only the case at hand. (i) Lowering the Riesz order from $m \geq 2$ to all $m > 0$ by the double Mellin transform technique of Brüdern, Kaczorowski and Perelli [6], and a converse: does $E_m(x) \ll x^{m-3/4+\varepsilon}$ imply the Riemann hypothesis for $\zeta_{\mathbb{Q}(\sqrt{D})}$?

A Computational programme

Every number in this paper comes from `followup/scripts/riesz-raw.ts` (exact Riesz means; seconds) and `followup/scripts/explicit-diag.py` (exponents, constants, residues by Cauchy integrals, comparison; about 25 minutes with primes to $4\cdot 10^6$ and ordinates to 100; residues cached). The following items are specified for execution elsewhere; none is needed for the theorems.

(C1) Table of $Q_{2,\rho}$, $Q_{3,\rho}$ for the first 8 zeros of $L(s,\chi_{-4})$ and 6 of ζ, with an independent check by direct differentiation of $\log G_m$ (analytic derivatives of the local factors) instead of Cauchy integrals. Input: `data/explicit-diag-terms-*.pkl`. Output: a LaTeX table. Minutes.

(C2) Zero sum to $\gamma \leq 300$ (`-analyse-only` reuses the cache; new poles about 2 hours), to see how the unexplained root mean square 0.29 decreases; a phase-randomised control (same ordinates, random phases) beside the shifted-ordinate control.

(C3) Other polynomials: $t^2 + t + 41$ ($D = -163$; the zero at 0.2029 treated as a known non-oscillating component), $t^2 - 2$ ($D = 8$), an S_3 cubic ($t^3 - 2$; zeros of $\zeta_{\mathbb{Q}(2^{1/3})}$ from the LMFDB, $\omega_f(p)$ from the factorisation of f modulo p). Requires generalising the driver from χ_{-4} to arbitrary $\omega_f(p)$. A day of programming, hours of computing per polynomial.

(C4) $\mathrm{Off}_f(H)$ for t^2+1 to $H = 10^8$ (extending `thesis/offdiag.ts`; a sum over d of two roots, cheap) and its spectrum in $u = \log H$ after removing the mean: presence or absence of lines at the $\mathrm{SL}_2(\mathbb{Z})$ Laplace parameters 9.5337, 12.1730, 13.7798, 14.3585, 16.1381 and at $\gamma/2 = 7.0673, 10.5110, 12.5054, \ldots$ (Section 5). Hours.

(C5) The function-field explicit formula for $f = t^2 - D$ over $\mathbb{F}_q[u]$, $q = 3, 5, 7$: L-polynomials, exponents (Table ??), residues on circles, comparison with `thesis/ff.ts`. A day.

(C6) Raw Riesz means to $x = 10^9$ by a segmented sieve of $A_f(n)$, extending the $\log x$ range from 8.3 to 11.5 and separating the ordinates 12.99 and 14.13. Hours, ≥ 16 GB or a segmented rewrite.

Data and code

`followup/scripts/riesz-raw.ts`, `followup/scripts/explicit-diag.py`, `followup/scripts/gen-wp0.py`; results in `followup/data/`. Part I's code is in `thesis/`.

Statement on authorship and the use of AI

The questions, the direction of the research and the decisions about scope were the first author's; the mathematics, the software and the text were produced by the Anthropic model Claude Fable 5.1 working interactively under the first author's direction, in September 2026; the first author has verified the code and the results and takes responsibility for the content. Most journals do not permit an AI system as an author; if the venue requires it, the second author line becomes an acknowledgement and this statement the required disclosure.

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